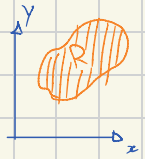
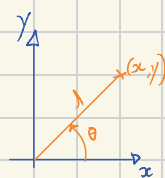


Lecture 11: Double integrals in polar coordinates; applications

$$\iint_R f(x,y) dA$$

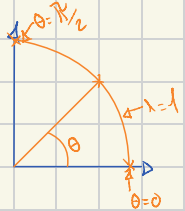


$$\iint_{x^2+y^2 \leq 1, x,y \geq 0} (1-x^2-y^2) dA = ? \text{ in polar coordinates?}$$



$$x = r \cos \theta$$

$$y = r \sin \theta$$

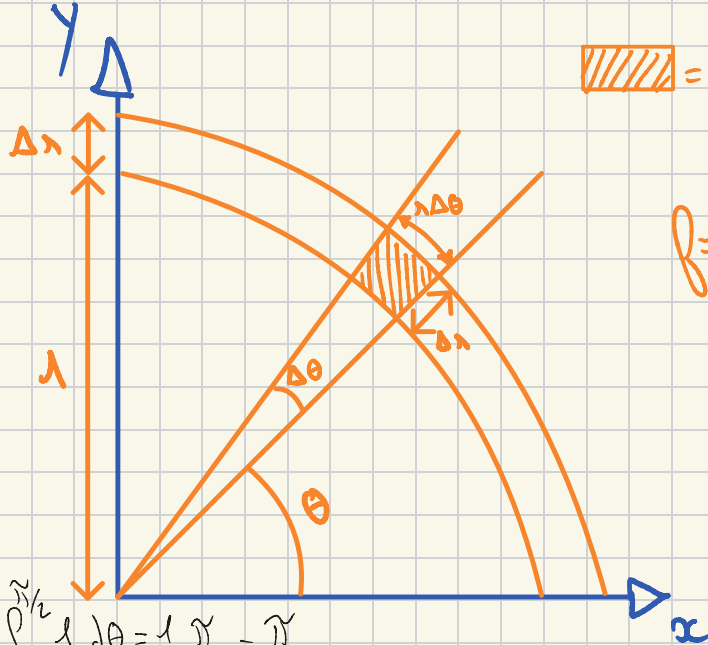


$$\int_0^{\pi/2} \int_0^1 f r dr d\theta$$

$$\Delta A \approx \Delta r \cdot r \Delta \theta$$

$$dA = r dr d\theta$$

$$f = 1 - x^2 - y^2 = 1 - (x^2 + y^2) = 1 - r^2$$



$$\int_0^{\pi/2} \int_0^1 (1-r^2) r dr d\theta = \int_0^{\pi/2} \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{1}{4} \frac{\pi}{2} = \frac{\pi}{8}$$

Applications: 1) Find area of Region R: $\text{area}(A) = \iint_R 1 dA$

Mass of a (flat) object with density δ = mass per unit area $\Delta m = \delta \cdot \Delta A$ Mass = $\iint_R \delta \cdot dA$

2) Average value of f in R:

$$\text{Average of } f = \bar{f} = \frac{1}{\text{Area}(R)} \iint_R f dA$$

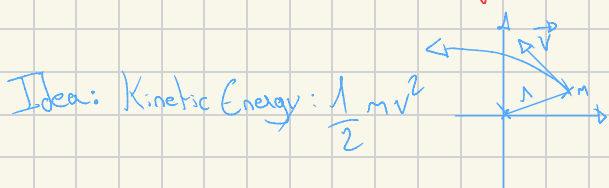
$$\text{Weighted average of } f \text{ with density } \delta: \frac{1}{\text{Mass}(R)} \iint_R f \delta dA$$

2a) Center of mass is located at $(\bar{x}, \bar{y})$ with $\bar{x} = \frac{1}{\text{Mass}} \iint_R x \delta dA$; $\bar{y} = \frac{1}{\text{Mass}} \iint_R y \delta dA$

3) Moment of inertia

Mass = How hard it is to impart translation motion (push/pull)

Moment of inertia: How hard it is to impart rotation motion about that axis



Idea: Kinetic Energy: $\frac{1}{2} m v^2$

$$\text{For a mass } m \text{ at distance } r \text{ and angular velocity } \omega. v = r \omega \Leftrightarrow E_c = \frac{1}{2} m v^2 = \frac{1}{2} m r^2 \omega^2$$

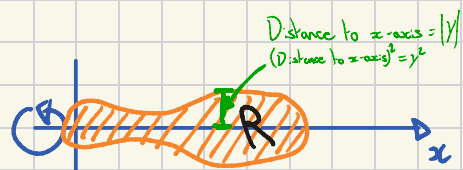
Coefficients of omega (ω) are the moment of inertia

For a solid with density δ : $\Delta m \approx \delta \Delta A$
has moment of inertia $\Delta m R^2 = r^2 \delta \Delta A$

$$\text{Moment of inertia about the origin: } \iint_R r^2 \delta dA = I_o$$

$$\text{Rotational kinetic energy is } \frac{1}{2} I_o \omega^2$$

What about: rotation over x-axis (Rotisserie Chicken)



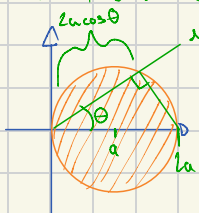
$$I_x = \iint_R y^2 \delta dA$$

Example: radius a δ density = 1
Rotate around its center



$$I_o = \iint_R r^2 dA = \int_0^{2\pi} \int_0^a r^2 r dr d\theta = 2\pi \frac{a^4}{4} = \frac{\pi a^4}{2}$$

Rotation around a point along its circumference:



$$I_o = \iint_R r^2 dA = \int_{-\pi/2}^{\pi/2} \int_0^{2a \cos \theta} r^2 r dr d\theta$$

$$I_{\text{center}} = \left[\frac{r^4}{4} \right]_0^{2a \cos \theta} = a^4 \cos^4 \theta \Leftrightarrow \int_{-\pi/2}^{\pi/2} a^4 \cos^4 \theta d\theta = \frac{3}{2} \pi a^4$$